

Teoría de las perturbaciones

Independiente del tiempo:

(Quantum Physics – Gasirowicz)

Caso no degenerado:

(Cada estado tiene su energía)

Consideremos un sistema descrito por el hamiltoniano H_0 :

$$H_0 \phi_n = E_n^0 \phi_n$$

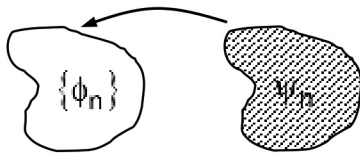
Se trata de encontrar, en forma aproximada, los valores propios y estados propios de:

$$H = H_0 + \lambda H_1, \quad \lambda \ll 1$$

Debemos resolver:

$$(H_0 + \lambda H_1) \psi_n = E_n \psi_n$$

$$\text{Dado que } \lambda \ll 1: \quad \begin{aligned} E_n &\rightarrow E_n^0 \\ \psi_n &\rightarrow \phi_n \end{aligned} \quad (*)$$



Como $\{\phi_n\}$ es una base, entonces:

$$\psi_n = N(\lambda) \left\{ \phi_n + \sum_{k \neq n} C_{nk}(\lambda) \phi_k \right\}$$

Considerando (*)

$$\begin{aligned} N(0) &= 1 \\ C_{nk}(0) &= 0 \end{aligned}$$

Expandamos en serie de potencias:

$$C_{nk}(\lambda) = \lambda \overset{\curvearrowright}{C}_{nk}^{(1)} + \lambda^2 C_{nk}^{(2)} + \dots$$

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots$$

$$(H_0 + \lambda H_1) \left\{ \phi_n + \sum_{k \neq n} \lambda C_{nk}^{(1)} \phi_k + \sum_{k \neq n} \lambda^2 C_{nk}^{(2)} \phi_k + \dots \right\} = E_n \{ \}$$

Identificando los coeficientes asociados a λ , obtenemos:

$$\text{Para } \lambda^0: H_0 \phi_n = E_n^{(0)} \phi_n$$

$$\text{Para } \lambda: H_1 \phi_n + H_0 \sum_{k \neq n} C_{nk}^{(1)} \phi_k$$

$$= E_n^{(1)} \phi_n + E_n^{(0)} \sum_{k \neq n} C_{nk}^{(1)} \phi_k \dots \quad (1)$$

Para λ^2 :

⋮

Consideremos $(\phi_k, \phi_\ell) = \delta_{k\ell}$

$$\text{De (1): } H_1 \phi_n + \sum_{k \neq n} (E_k^{(0)} - E_n^{(0)}) C_{nk}^{(1)} \phi_k = E_n^{(1)} \phi_n$$

$$C_{nk}^{(1)} = ?$$

$$E_n^{(1)} = ?$$

$$(\phi_n, E_n^{(1)} \phi_n) = (\phi_n, H_1 \phi_n)$$

$$+ \left(\phi_n, \sum_{k \neq n} (E_k^{(0)} - E_n^{(0)}) C_{nk}^{(1)} \phi_k \right) \quad (1)$$

$$\boxed{E_n^{(1)} = (\phi_n, H_1 \phi_n)} \quad \begin{array}{l} \phi_n, \phi_k \\ \text{ortogonales} \end{array}$$

Corrección en 1° orden de la energía

Si $m \neq n$:

$$(\phi_m, H_1 \phi_n) + \sum_{k \neq n} (E_k^{(0)} - E_n^{(0)}) C_{nk}^{(1)} (\phi_m, \phi_k) \\ = E_n^{(1)} (\phi_m, \phi_n) = 0$$

$$(\phi_m, H_1 \phi_n) + (E_m^{(0)} - E_n^{(0)}) C_{nm}^{(1)} = 0$$

$$\boxed{C_{nk}^{(1)} = \frac{(\phi_k, H_1 \phi_n)}{E_n^{(0)} - E_k^{(0)}}$$

Corrección en 1° orden para la función de estado.

Tarea: Determine $C_{nk}^{(2)}$ y $E_n^{(2)}$

$$E_n^{(2)} = \sum_{k \neq n} \frac{|(\phi_k, H_1 \phi_n)|^2}{E_n^{(0)} - E_k^{(0)}}$$

Ejem: $H = \frac{p^2}{2m} + \frac{m \omega^2 R^2}{2} + \lambda R;$

$$\underline{\lambda \ll 1}$$

$$\boxed{\begin{array}{l} A \phi_n = a \sqrt{n} \phi_{n-1} \\ A^\dagger \phi_n = a \sqrt{n+1} \phi_{n+1} \end{array}}; \quad a = \sqrt{\hbar \omega}$$

$$\left. \begin{array}{l} A = \frac{1}{\sqrt{2m}} P - i \omega \sqrt{\frac{m}{2}} R \\ A^\dagger = \frac{1}{\sqrt{2m}} P + i \omega \sqrt{\frac{m}{2}} R \end{array} \right\}$$

$$\Rightarrow R = \frac{1}{i\omega\sqrt{2m}}(A^\dagger - A)$$

$$\begin{aligned} R\phi_n &= \frac{1}{i\omega\sqrt{2m}}(A^\dagger\phi_n - A\phi_n) \\ &= \frac{\sqrt{\hbar\omega}\sqrt{n+1}}{i\omega\sqrt{2m}}\phi_{n+1} - \frac{\sqrt{\hbar\omega}\sqrt{n}}{i\omega\sqrt{2m}}\phi_{n-1} \\ &= \frac{\sqrt{\hbar}}{i}\frac{\sqrt{n+1}}{\sqrt{2m\omega}}\phi_{n+1} - \frac{\sqrt{\hbar}}{i}\frac{\sqrt{n}}{\sqrt{2m\omega}}\phi_{n-1} \end{aligned}$$

$$\Rightarrow (\phi_n, R\phi_n) = 0$$

$$\Rightarrow E_n^{(1)} = 0$$

$$\begin{aligned} E_n^{(2)} &= \sum_{k \neq n} \frac{|(\phi_k, H_1 \phi_n)|^2}{E_n - E_k} \\ &= \frac{\hbar(n+1)/2m\omega}{-\hbar\omega} + \frac{\hbar n/2m\omega}{\hbar\omega} \\ &= -\frac{1}{2m\omega^2} \end{aligned}$$

$$E_n = E_n^{(0)} - \frac{\lambda^2}{2m\omega^2}$$

$$E_n^{(0)} - E_k^{(0)} = (n-k)\hbar\omega$$

